Week 12 - Loss Function and Artificial Neural Network

Activity Questions Detailed Solutions

AQ 12.1

1) Given $h : \mathbb{R}^d \to +1, -1$ and $h(x) = sign(w^T x)$, which of the following is/are equivalent to $1(h(x) \neq y)$ *1 point*

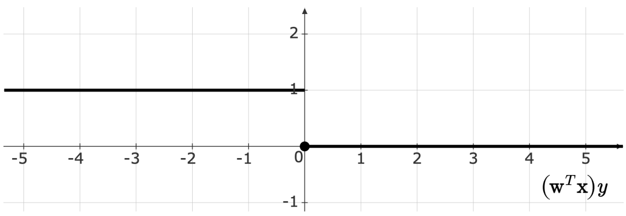
- $1((w^T x) * y \geq 0)$
- $1((w^T x) * y \leq 0)$
- \mathbb{1}((w^Tx)*y~\textgreater~0)
- \mathbb{1}((w^Tx)*y~\textless~0)

Solution

For a single training data-point (x, y), the $0-1$ loss can be given as:

$$\mathbb{I}\left[(\mathrm{w}^T\mathrm{x})y < 0\right]$$

Visually:



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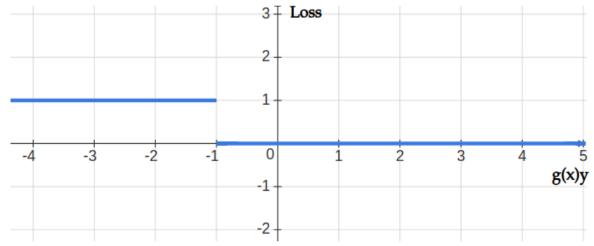
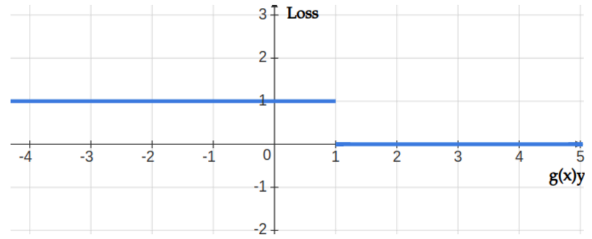
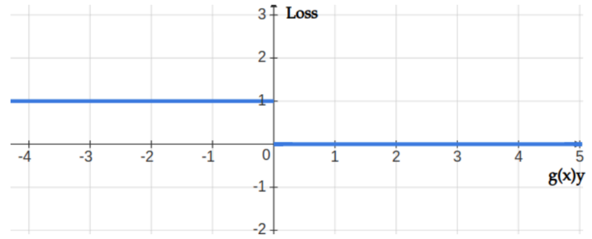
2) Assuming that a classification model produces an output $g(x)$ which is used to predict the value of y, which of the following plots represents 0-1 loss? *1 point*

○ 

○ 

○ 

Solution

(c)

3) Mark True or False: *1 point*

0-1 loss function is a continuous function, but not convex.

○ True

○ False

Solution

False, The 0-1 loss function is neither continuous nor convex. It is a discontinuous function because it jumps from 0 to 1 based on whether the prediction is correct or incorrect.

4) Mark True or False: *1 point*

Squared loss function is an appropriate surrogate loss function for 0-1 loss as it is continuous as well as convex.

○ True

○ False

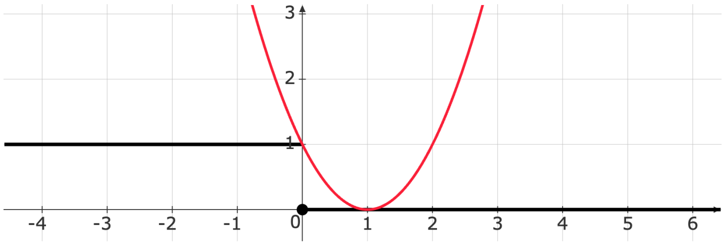
Solution

False, The squared loss function is not an appropriate surrogate loss function for 0-1 loss. While it is continuous and convex, it does not align well with the 0-1 loss in terms of optimization for classification tasks. The 0-1 loss is used in classification problems, where the goal is to minimize the classification error. Squared loss, on the other hand, is better suited for regression tasks and does not directly approximate the 0-1 loss effectively.

5) Squared loss L represented by $(g(x)y-1)^2$ may not be a good loss for a classification problem that produces $g(x)$ to predict $y \in +1, -1$ because *1 point*

- ○ L is not convex.
- ○ L can only be used for a linearly separable data.
- ○ If $sign(g(x))$ is not same as y, L will produce a zero loss.
- ○ Even if $sign(g(x))$ is same as y, L will produce a huge loss if $g(x)y \neq 1$

Solution



This loss is associated with a least squares classifier (inspired by regression). The loss is convex, but a poor approximation for the 0-1 loss. It heavily penalizes even points that are correctly classified, So, Even if sign(g(x)) is same as y, L will produce a huge loss if $g(x)y \neq 1$

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AQ12.2

1) Which of the following is true with respect to regularization in SVM loss function? *1 point*

- ○ Regularization strength is directly proportional to the value of $\mathcal{E}$.
- ○ Regularization strength is inversely proportional to the value of $\mathcal{E}$
- ○ Regularization strength is inversely proportional to the value of C.
- ○ Regularization strength is directly proportional to the value of C.

Solution

Regularization strength is inversely proportional to the value of C.

In Support Vector Machines, the regularization parameter C controls the trade-off between achieving a low training error and maintaining a large margin that helps generalization:

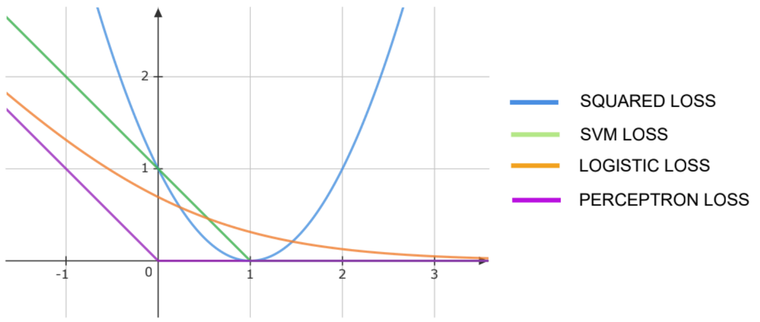
- A small value of C places more emphasis on maximizing the margin and allows for some misclassifications, which corresponds to stronger regularization. This helps prevent overfitting but may lead to underfitting.
- A large value of C prioritizes minimizing the training error by penalizing misclassifications heavily, which corresponds to weaker regularization. This can lead to overfitting if the model becomes too specific to the training data.

2) Which of the following is the best convex approximation of the 0-1 loss? *1 point*

- ○ Squared Loss
- ○ Absolute Loss
- ○ Hinge loss

Solution

Hinge Loss



3) Mark True or False: *1 point*

Hinge loss is a good surrogate function for 0-1 loss. Hence, the loss produced by the hinge loss will be exactly same as 0-1 loss for each data point.

- ○ True
- ○ False

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Solution

False, The loss produced by hinge loss is not exactly the same as the 0-1 loss for each data point. While hinge loss is a good surrogate for 0-1 loss due to its convexity and ability to approximate the behavior of 0-1 loss, it differs in how it penalizes predictions:

- 0-1 loss assigns a loss of 0 for correct predictions (where the sign of the prediction matches the label) and 1 for incorrect predictions, regardless of the margin or confidence.
- Hinge loss, on the other hand, penalizes predictions even if they are correct but fall within a certain margin (i.e., when $g(x)y < 1$). This means that hinge loss produces a non-zero penalty for correct predictions that are not confident enough.

4) In logistic regression, let S_1 be a scenario where labels (y) are represented as $0, 1$, and S_2 be a scenario where labels (y) are represented as $-1, +1$. What would be an appropriate expression of loss for a data point (x_i, y_i) in both the scenarios? *1 point*

Note: σ is the Sigmoid function.

- ○ $log(1 + e^{-w^T x_i y_i})$ in both the scenarios.
- ○ $S1 : log(1 + e^{-w^T x_i y_i})$,
  $S_2 : -y_i log(\sigma(w^T x_i)) - (1 - y_i) log(1 - \sigma(w^T x_i))$
- ○ $-y_i log(\sigma(w^T x_i)) - (1 - y_i) log(1 - \sigma(w^T x_i))$ in both the scenarios.
- ○ $S_1 : -y_i log(\sigma(w^T x_i)) - (1 - y_i) log(1 - \sigma(w^T x_i))$,
  $S_2 : log(1 + e^{-w^T x_i y_i})$

Solution

$$S_1 : -y_i log(\sigma(w^T x_i)) - (1 - y_i) log(1 - \sigma(w^T x_i)),$$
$$S_2 : log(1 + e^{-w^T x_i y_i})$$

AQ12.3

1) In the equivalent representation of Perceptron in terms of Stochastic Gradient Descent with modified Hinge Loss, what are the values of step size (η) and the gradient of loss function respectively? *1 point*

- 0, xy
- 1, xy
- 1, $-xy$
- 0, $-xy$

Solution

In the equivalent representation of the Perceptron algorithm as Stochastic Gradient Descent (SGD) with a modified hinge loss:

- The step size (η) is set to 1, as the Perceptron algorithm uses a constant step size for its updates.
- The gradient of the loss function is −xy, derived from the modified hinge loss $L = max(0 , -g(x)y)$. When ,$g(x)y < 0$ the gradient is computed as -xy , which corresponds to the direction of the update in the Perceptron algorithm.

2) Which of the following represents the loss function for Boosting? *1 point*

- $e^{-yh(x)}$
- $e^{yh(x)}$
- $e^{h(x)}$
- $e^{-h(x)}$

Solution

$$Loss(h , (x, y)) = e^{-y\, h(x)}$$

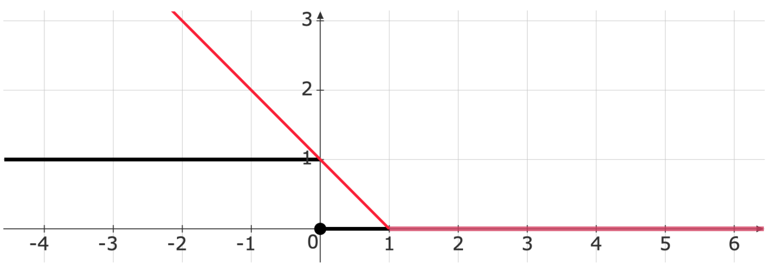
3) Mark True or False: *1 point*

Hinge Loss is convex but not differentiable.

- True
- False

Solution

True



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AQ12.4

1) State True or False: *1 point*

The loss function of neural network is a convex function.

- True
- False

Solution

False, The loss function of a neural network is not a convex function.

- Neural networks typically involve multiple layers of non-linear activation functions and numerous parameters (weights and biases). These non-linearities and the high-dimensional parameter space make the loss function highly non-convex.
- A non-convex loss function means that the optimization landscape contains multiple local minima, saddle points, and flat regions, making it challenging to find the global minimum.
- While optimization algorithms like stochastic gradient descent (SGD) and its variants are effective in practice, they do not guarantee convergence to the global minimum due to the non-convex nature of the loss function.

2) For a data set containing 10 features, how many neurons will the input and output layers respectively have in case of a regression problem? *1 point*

- 1, 10
- 10, 1
- 10, 10
- 1, 1

Solution

10,1

3) Which of the following may not be an appropriate choice for the activation function to be used for the hidden layers of a neural network? *1 point*

- Sigmoid function
- ReLU function
- 0-1 function

Solution

The 0-1 function is not an appropriate choice for the activation function in the hidden layers of a neural network due to the following reasons:

- Non-differentiability: The 0-1 function is not differentiable, which makes it unsuitable for optimization algorithms like backpropagation that rely on gradient computation.
- Binary output: The 0-1 function outputs discrete values (either 0 or 1), which limits its ability to represent complex patterns and relationships in data. Hidden layers require activation functions that introduce non-linearity and allow smooth transitions to capture intricate features.

4) The values -10, 20, 30, -30 will be transformed to which values using a ReLU function? *1 point*

- -10, 1, 1, -30
- 0, 1, 1, 0
- 0, 20, 30, 0
- -1, 20, 30, -1

Solution

$$a(z) = max\,(0\,,\ z)$$
$$a(-10\,,\ 20\,,\ 30\,,\ -30) = max\,(0\,,\ [\,-10\,,\ 20\,,\ 30\,,\ -30]) = [0\,,\ 20\,,30\,,\ 0]$$

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AQ12.5

1) If activation functions are not used in a neural network, it will be equivalent to *1 point*

- ○ SVM Regression
- ○ Naive Bayes Regression
- ○ Linear Regression

Solution

If activation functions are not used in a neural network, it will be equivalent to Linear Regression. Activation functions introduce non-linearity into the neural network, enabling it to model complex relationships and patterns in the data.

2) State True or False: *1 point*

Back-propagation is a technique of computing model parameters that always produces global minima of the loss function.

- ○ True
- ○ False

Solution

False, Backpropagation is a technique used to compute gradients of the loss function with respect to model parameters in a neural network, but it does not guarantee finding the global minimum of the loss function.

3) Which of the following loss functions are appropriate to be used at the output layer for (a) classification and (b) regression? *1 point*

- ○ (a) Squared loss, (b) Cross Entropy Loss
- ○ (a) Cross Entropy Loss, (b) Squared Loss
- ○ (a) Squared Loss, (b) Squared Loss
- ○ (b) Cross Entropy Loss, (b) Cross Entropy Loss

Solution

NN Loss function for Regression

$$L(NN\ ,\ (x\ ,\ \theta))\ =\ \sum_{i=1}^{n}(NN(x_i\ ,\ \theta)\ -\ y_i)^2$$

NN Loss function for Classification

$$CE\ =\ -\sum_{i=1}^{n} y_i\ log_2\ \widehat{y_i}$$

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